

Macroinvertebrate Ecology in Jervis Bay: Environmental Context, Pressures, Stakeholders and Management

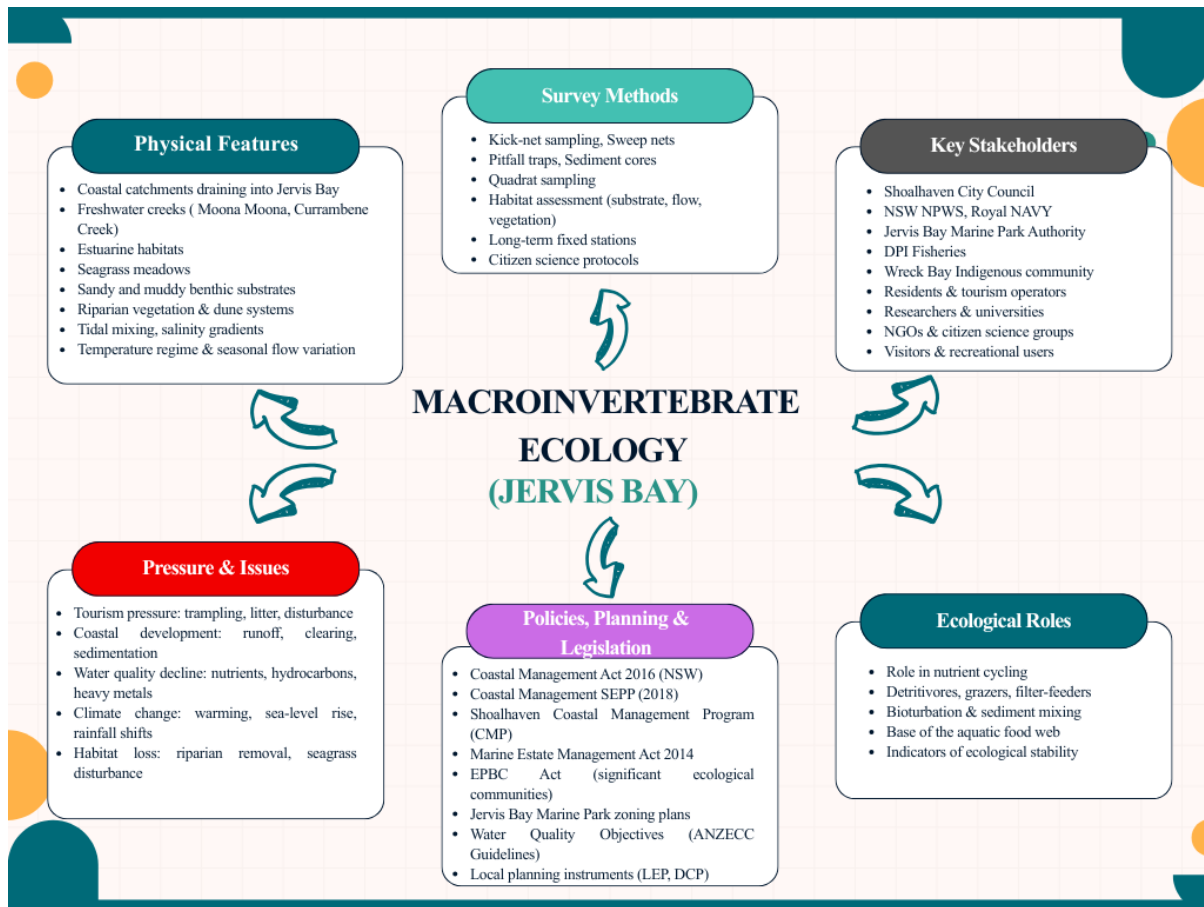


Figure 1: Mind Map on Macroinvertebrate Ecology in Jervis Bay

Macroinvertebrate communities are often used as ecological indicators due to their greater sensitivity to environmental change, their functional importance, and their ability to combine cumulative effects over time. The exposition outlines the physical characteristics of the Jervis Bay that impose on the ecology of the macroinvertebrates, the key anthropogenic pressures that influence these communities, the key stakeholder groups and the legislative and policy provisions that govern the management of these communities.

Physical Characteristics Relating to the Macroinvertebrate Ecology.

Jervis bay is a semi-enclosed embayment with clear waters, a variety of benthic habitats, and a blend of freshwater and marine. There are several freshwater creeks (such as Currumbene Creek and Moana Moona Creek) that supply freshwater to the bay, thus producing salinity gradients that affect the distribution of macroinvertebrates. The tributaries have sandy and muddy substrates, submerged vegetation, riparian vegetation, and dune systems which add to a complicated habitat matrix.

There are also large areas of seagrass meadows that are composed of *Posidonia* and *Zostera* species inhabiting the bay and are important nursery and food areas of the macroinvertebrates. The exchange of tidal energy with the surrounding ocean facilitates mixing of water masses hence influencing the sediment deposition and oxygenation. The seasonal changes in the rainfall affect the flow of freshwater thereby affecting the velocity of flow, turbidity, and nutrient supply. All these physical characteristics form a mosaic of habitats that support a wide range of macroinvertebrates.

Stresses and Problems that influence Macroinvertebrate Communities.

Macroinvertebrates are highly sensitive to environmental forces. One of the greatest causes of disruption in the Jervis Bay is tourism especially trampling of shorelines, recreational boating and litter production. Sediment is often stirred, so that when visitors are at their highest level, benthic habitats, upon which macroinvertebrates depend to seek shelter and forage, are disturbed.

There are other challenges of coastal development. The Shoalhaven region has been urbanised, and this has contributed to the amount of stormwater runoff into creeks and the bay. This runoff would add to water quality by introducing nutrients, hydrocarbons, sediments, and microplastics. There are numerous empirical data supporting the claim that pollution decreases the richness of macroinvertebrates and promotes the abundance of tolerant species (Duque et al., 2022). Sedimentation may smother the benthic habitats and inhibit oxygen in the estuarine systems.

These pressures are aggravated by climate change. An increase in water temperatures inevitably affects the metabolism and sometimes surpasses the limits of tolerance of vulnerable taxa. Rise in the sea level also helps in eroding dune systems and disruption of the riparian vegetation, and the modified patterns of rainfall help to regulate freshwater inflow and salinity. Such alterations in hydrology may alter the composition of macroinvertebrates community, which is observable in several catchments across the globe (Wang et al., 2023). In Jervis Bay, the change can cause the decline in the population of EPT taxa -groups known to be sensitive to warming and pollution.

Plastic and microplastic pollution also represent a new risk. Macroinvertebrates consume microplastics directly or through contaminated detritus and this further hinders growth and the flow of energy in the food web.

Key Stakeholders

Jervis Bay ecosystems involve the interaction of various stakeholder groups when it comes to the management of the ecosystems. Shoalhaven City Council is an important participant in the preparation and implementation of the Coastal Management Programs, stormwater regulation, and the control of developmental activities. The Jervis Bay national park and the surrounding reserves are managed by the NSW National Parks and Wildlife Service to administer the terrestrial and marine land use conservation.

Jervis Bay Marine Park Authority controls boating, fishing and conservation areas through the Marine Estate Management Act 2014. Local communities, especially the Wreck Bay

Aboriginal Community, have cultural and historical connections to the land and provide the invaluable ecological knowledge and stewardship.

Policies, Planning and Legislation.

Macroinvertebrate ecology is closely connected with environmental policy. Coastal Management Act 2016 (NSW) provides the general guidelines of the coastal hazard management, ecological resilience, and sustainable land use. Complementary Coastal Management SEPP (2018) delineates Coastal Environment Areas and Coastal Wetlands, thus specifying what can and cannot be done to vulnerable ecosystems.

The Shoalhaven Coastal Management Program (CMP) at the local scale determines the priority actions to rectify the water quality, stabilise the dune systems, reduce pollution, and improve the estuarine health. CMP activities directly affect recovery in macroinvertebrates since these species are cumulative on the pressures.

Marine estate management Act 2014 regulates marine parks, and zoning in the Jervis Bay marine park is included. Fishing, dredging and anchoring restrictions are used to conserve integrity of benthic habitats in sensitive areas.

At a national level, the EPBC Act could apply to the habitat of the threatened ecological communities and protection of habitat in the catchment of the bay. The ANZECC Guidelines also apply to water-quality monitoring, to which biological indicators such as macroinvertebrates are recommended in the determination of ecological condition.

Ecological Functions of Macroinvertebrates.

Macroinvertebrates are important in ecological processes. Detritivores break down organic matter thus enabling nutrition recycling. Filter feeders eliminate suspended particles, which enhances the clarity of the water. Burrowing life helps in the aeration of sediments, nutrient circulation and the degradation of pollutants (Kristensen et al., 2012). The combination of these processes ensures the quality of water and sustains greater trophic levels including fish and shorebirds.

Biodiversity is beneficial to ecosystems; more diverse communities maintain functional redundancy and stability during disturbance. It has been shown by empirical research that the loss of macroinvertebrate diversity undermines decomposition rates and ecosystem productivity (Cardinale et al., 2012).

Monitoring Methods

Monitoring and surveying cannot be done away with in understanding condition of the ecosystem. Long-term monitoring programmes include seasonal sampling, contributions by citizen-scientists, and fixed transects to identify change with time. It can be morphologically or genetically (DNA metabarcoding) identified and thus improve taxonomic accuracy. Analysis of data involves the use of measures like species richness, abundance, EPT richness, and SIGNAL2 scores, which are sensitive to pollution (Chessman, 2003). Multivariate analysis is used to determine spatial patterns and disturbance gradients.